

Intersection-Driven Enhanced Autonomous Safety (IDEAS) with LSTM Intention Prediction

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Abstract—Vehicle intention prediction is a pivotal aspect in the agility and safety of autonomous vehicles in all driving scenarios ; if genuine enhancement of autonomous vehicles are required , we need to make them adopt human interpretation of driver's intention especially in cases that require a lot of human interaction as well as complex driving behaviors like the ones at intersections, roundabouts and emergency cases such as sudden stops where vehicle intention prediction helps in taking the correct evasive action within a real time period where every second of action makes an impact and can prevent a catastrophe from taking place .In the worst case, it helps minimize the damage and make safety a priority. Intention prediction can also be used to enhance trajectory prediction(intention conditioned trajectory prediction). In this study, an LSTM model is proposed to predict the vehicle's intention at intersections 2 s before the event takes place such that the intentions predicted are going straight , turning left and turning right. Various model experiments and parameters are thoroughly tested and explained in the experiments section . InD dataset was used but our own pipeline was developed (IDEAS).

Index Terms— Autonomous Vehicles, Intelligent vehicles ,Intention prediction ,Intersections,Behaviour prediction, LSTM ,deep learning models, dataset labeling.

I. INTRODUCTION

In continuous efforts of enhancing road safety to reduce accidents , autonomous vehicles could be a leading participant if it is able to avoid main causes of accidents like human error, having a misleading intention , taking the wrong action , unexpected lane changes, vague driving behaviors at intersections and sometimes not seeing vehicles if they are occluded.Understanding the scene and even predicting the intention of the target vehicle could alert the driver if the driven car is not fully autonomous or even better, if it is fully autonomous, would take the safest evasive action to ensure optimal safety. Intention prediction is vital for all road users for example: pedestrians,cyclists and vehicles.

For each road user, there are even various traffic and road scenarios like pedestrian intention prediction at intersections and highways .In this paper ,vehicle intention prediction is going to be our main focus. Car accidents at intersections suffer fatality and severity for few reasons like:vehicles could be at high speed, the collision angle could be deadly (90 degrees) ; imagine collision of 2 cars one that is crossing

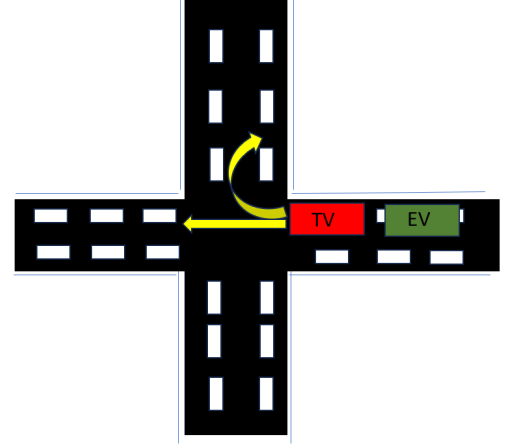


Fig. 1: An illustration of possible driving maneuvers of the target vehicle where the target vehicle is the red one and the ego vehicle is the green one.

from South to North and another one from East to West) which reinforces the vitality of addressing the challenge of vehicle intention prediction at intersections which could exceedingly improve road safety and accelerate the development process of autonomous vehicle. This is why vehicle intention prediction at intersections is our paper motivation.

This paper is organized in the following sections: Section II is going to discuss Related Work. Section III is going to discuss the IDEAS pipeline .Section III is going to discuss the features of our deep learning model . Section IV is going to discuss experiments and results. Lastly, Section V is going to discuss conclusion and future recommendations.

II. RELATED WORK

Vehicle intention prediction is a dominant factor in the agility and safety of autonomous vehicles .Intention prediction is a classification problem thus diverse classifiers throughout the years have been exploited to carry out intention prediction where some are categorised as machine learning techniques such as support vector machines (SVM) [1] and hidden Markov models (HMM) [2].

The other category is deep learning models , that have been proven exceedingly significant such as Recurrent Neural Networks (RNN) which are best utilised to model short sequences of data but are deficient in modelling long sequences due to the vanishing gradient problem , and that's why Long Short Term Memory (LSTM) is predominantly used in using the past sequential trajectory info of a vehicle to be able

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to predict the intention overcoming the hindrance of long sequences [3].

Additionally, a lot of recent studies have been speculating different variants of LSTM in vehicle intention prediction as in [4] where bidirectional LSTM coupled with attention mechanisms are used.

III. IDEAS PIPELINE IMPLEMENTATION

A. IDEAS Pipeline Overview

Check Fig. 2 for detailed pipeline overview.

B. The Data Preparation

As far as we are concerned and from the thorough search we carried out on a long period of time a huge research gap that we faced is that there was not any ego vehicle view dataset capturing driving behaviours at intersections so we decided to work with BEV datasets and leverage the advantages of not being constrained by sensor occlusions or limited ranges. Not only was it decided to use BEV, but also it was planned that the model that will be built would rely on features that could be extracted regardless the view thus it would be general enough and view independent thus easily applicable on multiple views and scenarios.

InD [5] which is a BEV dataset capturing different driving behaviours at different intersection structures (T intersection, cross-intersection, etc..) as in Fig. 3 was used.

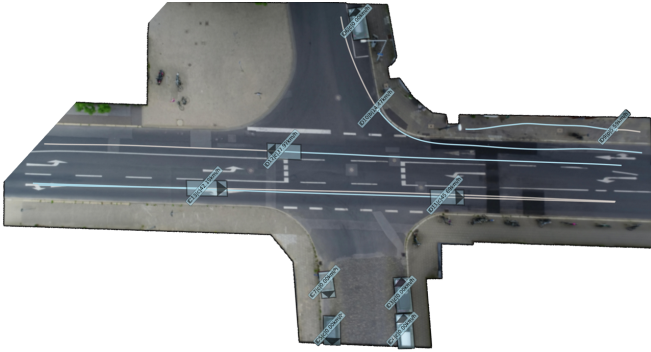


Fig. 3: Visualizer tool offered by the inD dataset to observe vehicle behaviour at different frames

Unfortunately, a problem was faced that no ground truths were available alongside the dataset trajectories so labelling had to be done thus our own developed pipeline (IDEAS) was instantiated as mentioned in the upcoming steps:

- **Step 1:** Using open-drive maps that were included in the dataset as shown in Fig. 4 when opened (by online open drive viewer), through moving the cursor over the maps, the x and y coordinates are yielded thus manual coordinate extraction was done for every road. As a preliminary step, all lower and upper limits of x and y coordinates of all roads were coded in order to label the road the vehicle is in at each frame based on the x and y coordinates of the vehicle's position. Solely using x and y coordinates, common areas between roads were found so another property was needed in order to distinguish

between roads leading to the inclusion of the heading angle which successfully identified between roads.

Common areas between roads still existed even after heading inclusion so a common range was extracted and the road column value was coded as "either road x or road y" till the vehicle moved to a unique range linked to a single road.

A question could come to mind and the question might be how is the heading angle extracted when all the open-drive map offers are the x and y coordinates? Using the visualizer tool, for example, if we wanted to extract the heading of a certain road, we observed different vehicles that drove in the same road and getting their ids, we refer to the csv file and observe their heading angles and extract a common range that fits all vehicles driving on each road.

The inclusion of the heading angle was done for all roads not just the ones that had overlaps. There was an unknown value filled in the column of road if something did not fit the range or a vehicle was not obeying the lane markings or allowed directions.

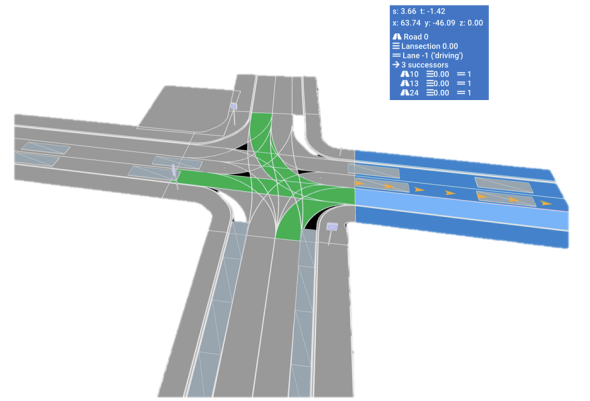


Fig. 4: a cross intersection open-drive map to extract the x and y coordinates of different roads

- **Step 2:** After the road column was filled for each frame, road transitions were coded to label events / ground truth behaviour as either going straight, turn left or turn right. Afterwards, it was manually validated and revised against the visualizer tool to make sure the ground truth matches the real action the vehicle took.
- **Step 3:** Furthermore, another filtration algorithm was written to filter out and exclude tracked objects of classes pedestrians, parked vehicles/ stationary vehicles and cyclists in order to merely focus on vehicle intention prediction for moving vehicles.

C. Features

Below are the main features used: the vehicle's center positions and heading angle $[P_x, P_y, \theta]$, vehicle velocity and acceleration $[v_x, v_y, a_x, a_y]$, longitudinal and lateral velocity and acceleration $[v_{lon}, v_{lat}, a_{lon}, a_{lat}]$, and the vehicle dimensions including the length l and width w .

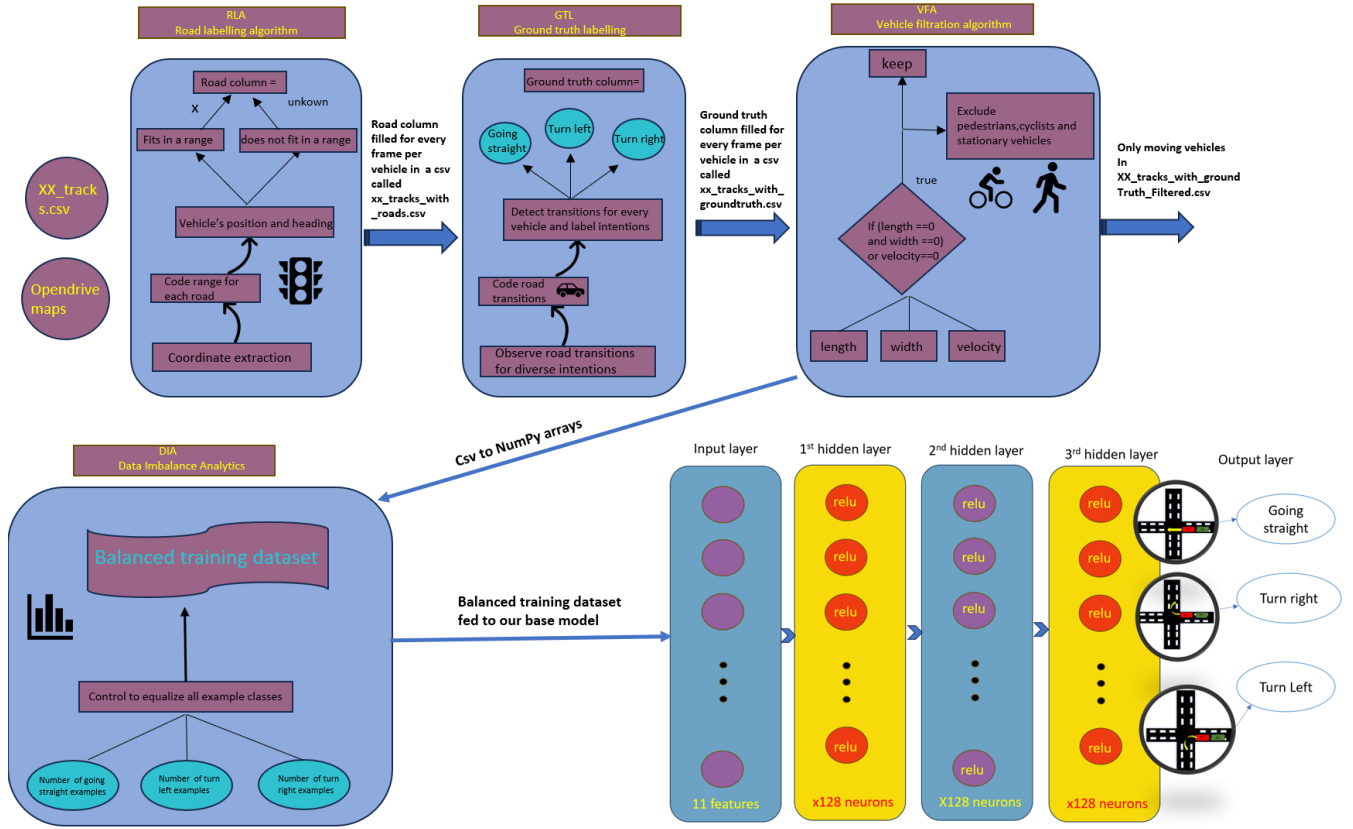


Fig. 2: The IDEAS Pipeline Stages Overview

In the experiments section, diverse models are experimented with a subset of these features and the results' comparisons are discussed.

D. Model

1) *Specs and tools:* Our model proposal is an LSTM deep learning model in order to exploit sequential nature of the frames and numerical features of each vehicle. Tensorflow and keras api were used and our model was coded in visual studio code. The hardware used to run the model is a laptop with an i9-14900HX cpu and nvidia's geforce rtx 4080. The intention prediction of a vehicle is treated as a classification problem where the output consists of 3 classes which are going straight, turn left and turn right.

2) *Input preparation:* For the inputs a subset of the features mentioned in the aforementioned features section were used, but initially they were in the form of a csv, so an input preparation algorithm was written to transform filtered csv (csv excluding pedestrians, cyclists and stationary vehicles) into 3d numpy arrays to be fed to the model along with a one dimensional labels array (from the ground truth column mentioned earlier) to be input as the ground truth of the model.

3) *Controlling imbalance:* Using pandas, an analytical part was run to investigate the number of input examples from each class from going straight, turn right and turn left. The justification behind the analytics part is to avoid data imbalance. Our doubts were validated and the going

straight input examples were exceeding the other 2 classes so a counter was used and only a number equal to the examples of the other 2 classes (turn right and turn left) were input as examples of going straight.

4) *Details of input reshaping:* InD dataset has a frame rate of 25 frames per second so in our model a 20 frame window size was used and padding was done for vehicles that had less than 20 frames before the event happened. Our model predicts the intention of the target vehicle 2s before the action occurs so as per the frame rate mentioned above, in data preparation an algorithm was written such that for every vehicle in the filtered csv, it gets the row number where the action happens and goes back 50 rows (1 row = 1 frame) and gets the row number of the first frame for each vehicle and then does feed sequences of 20 frames (aligned with the ground truth in the one dimensional labels array) as a single example till it reaches the end row (which is the row the action happened - 50 rows).

5) *Model architecture:* An input layer with a combination of the features mentioned in the features section was used and further details are to be further discussed in the experiments section. Then our model consists of 3 hidden layers each layer has 128 neurons and all 3 hidden layers use the RELU activation function as our base model. Finally, the output layer has 3 output nodes and uses soft-max function in order to limit output between 0 and 1 acting as a probability like one where the class with the highest probability is considered the output belonging to this class/intention in our case.

Batch Normalisation is used between the hidden layers. As kpis/evaluation metrics accuracy , precision recall,f1-score and loss are used. Our dataset is split as 80 percent training data, 20 percent validation data and for test data set a separate recording at the same location was used to further evaluate the model.

The categorical cross entropy was used as a loss function and as an optimization function Adam optimizer was used with a learning rate of 0.001. A batch size of 32 and 50 epochs were used.

IV. EXPERIMENTS AND RESULTS

In this section , model experiments that have been speculated is going to be clearly discussed ,categorised and their results are going to be compared respectively on training, validation and test datasets. Additionally, the performance graphs are going to be attached for some of the experiments for visualising the results.

A. Experiment 1: Testing different features

The main aim of this experiment is to test the efficacy of using different number of input features on our base model and on the model accuracy, precision, recall, F1-score and loss. In this experiment, 4 different variations have been created using 5, 7, 11, and 13 features respectively. The input features of the models respectively are $[P_x, P_y, \theta, v_x, v_y]$, $[P_x, P_y, \theta, v_x, v_y, a_x, a_y]$, $[P_x, P_y, \theta, v_x, v_y, a_x, a_y, v_{lon}, v_{lat}, a_{lon}, a_{lat}]$ and $[P_x, P_y, \theta, v_x, v_y, a_x, a_y, v_{lon}, v_{lat}, a_{lon}, a_{lat}, l, w]$.

The Architecture used for the 4 variations is unified for the sake of the comparison and any different parameter would be solely mentioned in each experiment accordingly. The unified architecture is an LSTM model with 3 hidden layers using relu activation functions and an output layer which uses softmax activation function for 3 output nodes which are going straight, turn left and turn right .The optimizer used is Adam optimizer with the categorical cross entropy loss function and a learning rate of 0.001. Batch normalization was used. The batch size is set to 32 and epochs to 50 where our sequence length is 20 frames ,our step size is 12 frames and our prediction window is 2s before the event occurs.

Table I illustrates that the 11 feature model yielded the best outcome thus we proceed with our upcoming experiments applying all variations on the model but in reference with the 11 feature model as our base model and comparing every variant with the base model.

B. Experiment 2: Other model variants

In the upcoming experiments, all model details of the base model are invariant and any modification will be solely mentioned in each experiment respectively.

1) *Adding regularization layers:* Regularization and dropout layers of parameter 0.1 are added to help reduce over-fitting if there exists any yielding a maximum training accuracy of 0.9540, Precision of 0.9559, Recall of 0.9527, and loss of 0.2758. The maximum validation accuracy is

0.9971, val Precision is 1.0000, val Recall is 0.9912, and val loss is 0.4245. Fig. 5 shows graph results.

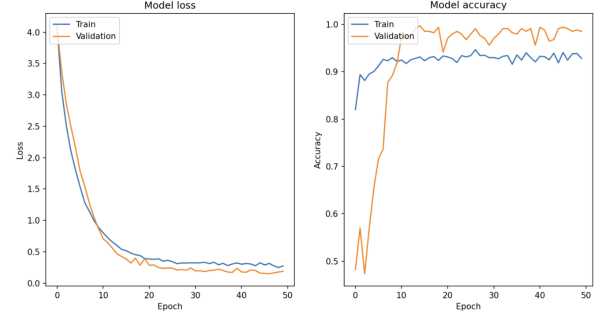


Fig. 5: accuracy and loss of training and validation datasets across epochs of the regularized model

2) *Bidirectional LSTM:* The first hidden layer of the base model is made bidirectional to testify the effect of a bidirectional model and its impact on the results. A maximum training accuracy of 0.9727 ,Precision of 0.9727 Recall of 0.9725 and loss of 0.0795 .The maximum validation accuracy is 0.9971 , val Precision is 0.9971 ,val Recall is 0.99712 and val loss is 0.0266 were yielded. Fig. 6 shows graph results.

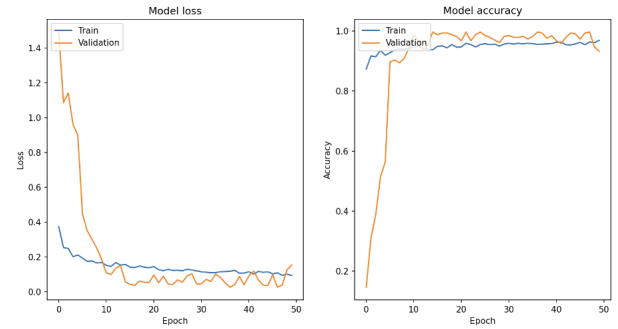


Fig. 6: accuracy and loss of training and validation datasets across epochs for bidirectional LSTM

TABLE IV: Results of bidirectional LSTM on test dataset

Class	Precision	Recall	F1-score	support
Going Straight	0.78	0.96	0.86	719
Turn Left	0.94	0.79	0.85	857
Turn Right	0.82	0.78	0.80	486

3) *Trying different activation functions:* The 3 hidden layers of the LSTM is set once as sigmoid and once as tanh to testify the influence of hidden layers' activation function choice on results. Our base model which used relu has the highest output , the the tanh and lastly the sigmoid has the lowest output but they are all slight differences(1 percent at maximum). Fig. V and VI show graph results.

TABLE I: Results of Experiment 1 comparing the 4 different models

Metric	5-Feature Model	7-Feature Model	11-Feature Model	13-Feature Model
Test Accuracy	0.8264	0.87	0.9151	0.8288
Test Precision	0.8268	0.8705	0.9151	0.8331
Test Recall	0.8264	0.8700	0.9146	0.8278
Test F1 Score	0.8254	0.8698	0.9150	0.8294
Test Loss	0.5811	0.7418	0.7214	1.4496

TABLE II: Results of Experiment 2 comparing all model variants with respect to the base model

Metric	base model	Regularization	bidirectional	25 window	2 hidden layers	4 hidden layers	0.01 learning rate	1.5s prediction window
Test Accuracy	0.9151	0.9146	0.8443	0.8923	0.8909	0.8555	0.5587	0.8749
Test Precision	0.9151	0.9160	0.8457	0.8941	0.8917	0.8570	0.5650	0.8766
Test Recall	0.9146	0.9146	0.8429	0.8923	0.8904	0.8545	0.5291	0.8715
Test F1 Score	0.9150	0.9142	0.8438	0.8930	0.8906	0.8554	0.5246	0.8746
Test Loss	0.7214	0.5267	0.9669	0.6291	1.8022	0.9263	40604.2773	1.1998

TABLE III: Results of regularized model on test dataset

Class	Precision	Recall	F1-score	support
Going Straight	0.88	0.95	0.91	719
Turn Left	0.94	0.93	0.94	857
Turn Right	0.94	0.83	0.88	486

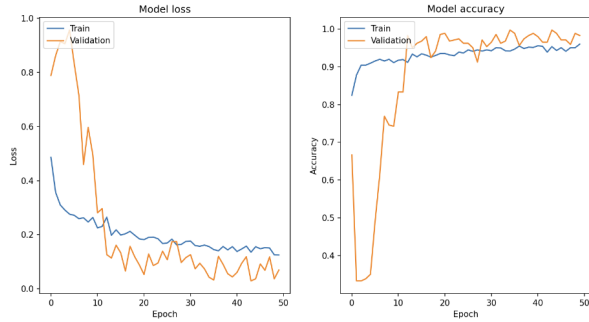


Fig. 7: accuracy and loss of training and validation datasets across epochs of using sigmoid as an activation function

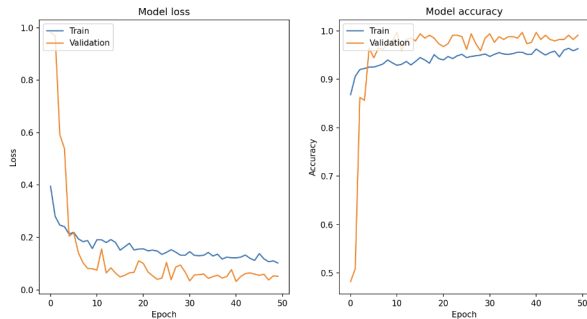


Fig. 8: accuracy and loss of training and validation datasets across epochs of using tanh as an activation function

4) *Trying different window sizes/sequence length:* In this paper, we tried 2 window sizes which are 20 (base model) and 25. By comparing results of the test datasets, it is shown

TABLE V: Results of using sigmoid as an activation function on test dataset

Class	Precision	Recall	F1-score	support
Going Straight	0.86	0.95	0.90	719
Turn Left	0.93	0.92	0.92	857
Turn Right	0.96	0.82	0.88	486

TABLE VI: Results of using tanh as an activation function on test dataset

Class	Precision	Recall	F1-score	support
Going Straight	0.86	0.96	0.91	719
Turn Left	0.96	0.91	0.93	857
Turn Right	0.93	0.84	0.89	486

that the window size of 20 showed better results than window size of 25. Fig. VII shows the graph results.

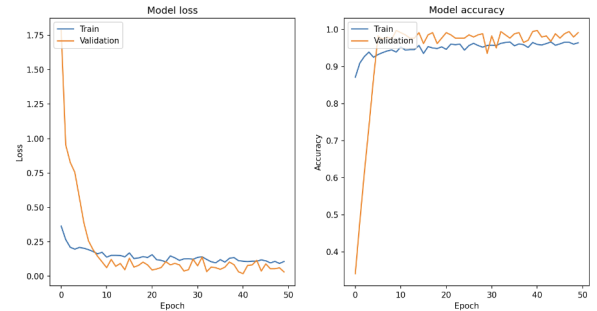


Fig. 9: accuracy and loss of training and validation datasets across epochs of using a window size =25

TABLE VII: Results of using a window size =25

Class	Precision	Recall	F1-score	support
Going Straight	0.81	0.97	0.88	719
Turn Left	0.96	0.85	0.90	857
Turn Right	0.94	0.85	0.89	486

5) *Ablation studies: trying different number of hidden layers:* In relevance to our basic model which had 3 hidden

layers where they all have relu activation functions , we investigate 2 hidden layers and 4 hidden layers thus we once try removing a layer and once adding a layer. The results stated that our base model is the best , then the 2 hidden layers and lastly the 4 hidden layers.

TABLE VIII: Results of using only 2 hidden layers

Class	Precision	Recall	F1-score	support
Going Straight	0.87	0.97	0.92	719
Turn Left	0.96	0.84	0.89	857
Turn Right	0.83	0.86	0.84	486

TABLE IX: Results of using 4 hidden layers

Class	Precision	Recall	F1-score	support
Going Straight	0.84	0.92	0.88	719
Turn Left	0.90	0.81	0.86	857
Turn Right	0.80	0.84	0.82	486

6) *Tackling different learning rates:* A learning rate of 0.01 is tested and by comparison of test data set results , the 0.001 learning rate (base model)yielded way better results than the 0.01 .

TABLE X: Results of using an 0.01 learning rate

Class	Precision	Recall	F1-score	support
Going Straight	0.96	0.22	0.36	719
Turn Left	0.62	0.62	0.62	857
Turn Right	0.45	0.96	0.61	486

7) *Prediction window:* In our base model , the predictions were 2 seconds earlier than the real event (base model) but we also opted to try the prediction only 1.5 seconds earlier.The outcome is that predicting 2 s earlier yielded better results.

TABLE XI: Results of using a 1.5 sec prediction window

Class	Precision	Recall	F1-score	support
Going Straight	0.85	0.97	0.91	719
Turn Left	0.95	0.82	0.88	857
Turn Right	0.80	0.83	0.81	486

It was witnessed in the aforementioned variations that tuning a single parameter exceedingly affects the model , so to reach the best model plenty of trial and error has to be carried and in our experiments the base model is the best one , but this should not hold true for different data sets.Further more, it was alternating a single factor and measuring the performance but also other maximal performances could be obtained if more than one factor are changed at a time.

V. CONCLUSION AND FUTURE RECOMMENDATIONS

In this paper, LSTM has been proven exceedingly effective and promising in solving the classification problem of intention prediction and the most outstanding model in our work is the base model yielding a maximum training accuracy of 97.24 percent and a maximum validation accuracy of 99.71 percent .

As future recommendations and in extending efforts of this paper, it is aimed that our model is utilized in intersection management and further tested on other environments and driving scenarios.Additionally,it is planned to experiment different deep learning architectures like transformers which has been shown to be effective in similar areas of research.

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